La Scuola d'Italia / Dr. Huson / IB Math 11 (a)
28 October 2026

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2.6 Quiz: Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. [Maximum mark: 10]

   Given the points $A(-4,\ 1)$, $B(4,\ 5)$ and $C(1,\ 6)$.

   (a) Find the gradient of the line AB. [2]

   (b) Find the equation of the line AB. [2]

   The line L passes through C and is perpendicular to the line AB.

   (c) Find the equation of L. [3]

   (d) Find the coordinates of the point where L meets the line AB. [3]

(a) $m_{AB} = \dfrac{5-1}{4-(-4)} = \dfrac{1}{2}$

(b) $A(-4, 1)$, $m = \frac{1}{2}$

$\overleftrightarrow{AB}$ $\quad y - 1 = \frac{1}{2}(x+4)$

(c) $C(1, 6)$, $m_\perp = -2$

L: $\quad y - 6 = -2(x-1)$

(d) $\frac{1}{2}(x+4) + 1 = -2(x-1) + 6$

$\frac{5}{2}x = 5$

$x = 2$

$y = \frac{1}{2}(2+4) + 1 = 4$

$(2, 4)$

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2. [Maximum mark: 7]

At a cinema, an adult ticket costs x euros and a child ticket costs y euros. A group of 2 adults and 3 children pays 47 euros. A group of 1 adult and 4 children pays 41 euros.

(a) Write down two equations in x and y. [2]

(b) Find the price of an adult ticket and the price of a child ticket. [3]

(c) A family of 2 adults and some children pays 61 euros. Find the number of children in the family. [2]

(a) $2x + 3y = 47$ (1)
$1x + 4y = 41$ (2)

(b) $2x + 8y = 82$ $2\times(2)$
$2x + 3y = 47$ (1)

$0x + 5y = 35$ $[2\times(2)]-(1)$
$y = 7$ child

$2x + 3(7) = 47$
$x = 13$ adult

(c) $2(13) + n(7) = 61$
$n = 5$ children

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3. [Maximum mark: 10]

(a) Simplify $(2a^3b)^2 \times 3a^{-2}b^{-4}$. Give your answer using positive exponents only. [3]

(b) Find the exact value of $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$. [2]

(c) Solve the equation $9^{x-1} = 27^{x-3}$. [3]

(d) Solve the equation $2^{x+3} = \frac{1}{32}$. [2]

(a) $(4 \times 3)\, a^{6-2} b^{2-4} = 12 a^4 b^{-2}$

$= \frac{12a^4}{b^2}$

(b) $\left(\frac{\sqrt[3]{27}}{\sqrt[3]{8}}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

(c) $3^{2(x-1)} = 3^{3(x-3)}$

$2(x-1) = 3(x-3)$

$x = 7$

(d) $2^{x+3} = 2^{-5}$

$x+3 = -5$

$x = -8$

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4. [Maximum mark: 8]

In this question, give every answer in exact form.

(a) Simplify $\sqrt{50}+\sqrt{8}-\sqrt{32}$. [2]

(b) Expand and simplify $\left(3+\sqrt{5}\right)\left(3-\sqrt{5}\right)$. [2]

(c) Hence, write $\dfrac{8}{3-\sqrt{5}}$ in the form $a+b\sqrt{5}$, where $a, b \in \mathbb{Z}$. [2]

(d) Solve the equation $x\sqrt{2}+4=3x\sqrt{2}$. Give your answer in its simplest form. [2]

(a) $5\sqrt{2}+2\sqrt{2}-4\sqrt{2}$

$= 3\sqrt{2}$

(b) $9-(\sqrt{5})^2 = 4$

(c) $\dfrac{8}{3-\sqrt{5}} \cdot \dfrac{3+\sqrt{5}}{3+\sqrt{5}} = \dfrac{24+8\sqrt{5}}{4}$

$= 6+2\sqrt{5}$

(d) $x\sqrt{2}+4 = 3x\sqrt{2}$

$4 = 2x\sqrt{2}$

$x = \sqrt{2}$

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2.6 Quiz: Lines, Exponents and Radicals

A calculator is allowed. When a question asks for an exact answer or a particular form, show the algebra that leads to it.

1. [Maximum mark: 12]

   The line L_1 has equation $3x + 4y = 24$. It meets the x-axis at the point A and the y-axis at the point B. The point P has coordinates $(-1, -3)$.

   (a) Find the gradient of L_1. [2]

   The line L_2 passes through P and is perpendicular to L_1.

   (b) Find the equation of L_2. [3]

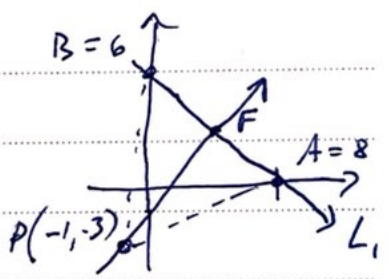
   (c) Find the coordinates of the point F where L_1 and L_2 intersect. [3]

   (d) Find the area of triangle PAB. [4]

(a) L_1: $3x+4y=24$

$y = -\frac{3}{4}x + 6$

$m = -\frac{3}{4}$



(b) $m_\perp = \frac{4}{3}$

L_2: $y+3 = \frac{4}{3}(x+1)$

(c) $-\frac{3}{4}x + 6 = \frac{4}{3}(x+1) - 3$

$-\frac{9}{12}x + 6 = \frac{16}{12}x - \frac{5}{3}$

$92 = 25x$

$x = \frac{92}{25}$

$y = -\frac{3}{4}\left(\frac{92}{25}\right) + 6 = -\frac{276}{100} + 600$

$= \frac{81}{25}$

(d) $AB = \sqrt{8^2+6^2} = 10$

$PF = \sqrt{\left(\frac{92}{25}+1\right)^2 + \left(\frac{81}{25}+3\right)^2} = \frac{39}{5}$

$A = \frac{1}{2}(10)\left(\frac{39}{5}\right) = 39$

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2. [Maximum mark: 7]

Solve the following system of equations by elimination. Show your working. [7]

$$2x + 3y - 4z = -20 \quad (1)$$
$$5x - 2y + 3z = 34 \quad (2)$$
$$3x + 4y + 2z = 11 \quad (3)$$

$26x + 1y = 76$ } $3.(1) + 4.(2)$; (4)

$8x + 11y = 2$ } $(1) + 2\times(3)$; (5)

$278x + 0y = 834$ $11.(4) - (5)$

$x = 3$

$26(3) + y = 76$

$y = -2$

(3): $3(3) + 4(-2) + 2z = 11$

$z = 5$

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2.6 Early Finishers: Lines, Exponents, Radicals and Sequences

For students who finish the quiz early. A calculator is allowed.

1. [Maximum mark: 11]

   (a) Solve the equation $25^{x+1} = 125^{2x-3}$. [3]

   (b) Find the integers a and b such that $\left(2+\sqrt{3}\right)\left(a+b\sqrt{3}\right) = 12+7\sqrt{3}$. [3]

   Consider the equation $3\left(9^x\right) - 82\left(3^x\right) + 27 = 0$.

   (c) By letting $u = 3^x$, solve the equation. [5]

(a) $5^{2(x+1)} = 5^{3(2x-3)}$

$2(x+1) = 3(2x-3)$

$11 = 4x$

$x = \frac{11}{4}$

(b) $2a + 2b\sqrt{3} + a\sqrt{3} + b(\sqrt{3})^2 = 12 + 7\sqrt{3}$

$(2a+3b) + (a+2b)\sqrt{3} = 12 + 7\sqrt{3}$

$2a+3b=12$ (1)

$a+2b=7$ (2)

$0a+1b=2$ $\quad 2\times(2)-(1)$

$a=3,\ b=2$

(c) $3(9^x) - 82(3^x) + 27 = 0 \quad u = 3^x$

$3u^2 - 82u + 27 = 0$

$(3u-1)(u-27) = 0$

$u = \frac{1}{3},\ u = 27$

$x = -1,\ 3$

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2. [Maximum mark: 8]

An arithmetic sequence has first term $u_1 = 7$ and common difference 3. A geometric sequence has first term $v_1 = 2$ and common ratio 1.5.

(a) Write down an expression for u_n and an expression for v_n. [2]

(b) Find the least value of n for which $v_n > u_n$. [3]

(c) Find the least value of n for which the sum of the first n terms of the geometric sequence is greater than the sum of the first n terms of the arithmetic sequence. [3]

(a) $u_n = 7 + 3(n-1)$

$v_n = 2 \cdot (1.5)^{n-1}$

(b) $v_n = 2 \cdot (1.5)^{n-1} > 7 + 3(n-1) = u_n$

n	u_n	v_n
7	25	22.8
8	28	34.2

$n = 8$

(c) $S_{u,n} = \frac{n}{2}(2 \cdot 7 + 3(n-1))$

$S_{v,n} = \frac{2(1.5^n - 1)}{1.5 - 1} = 4 \cdot 1.5^n - 4$

n	$S_{u,n}$	$S_{v,n}$
9	171	149.8
10	205	226.7

$n = 10$